Respiratory-Related Emergency Department Visits

Technical Report

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Background and Research Questions

To understand our question, it is important to first explain a few niche variables. PM stands for Particulate Matter, which are airborne particles small enough to be inhaled into the respiratory tract. The numbers refer to the size: PM2.5 particles have diameters <= 2.5 micrometers, while PM10 particles are <= 10 micrometers. These are measured in micrograms per cubic meter of air ($\mu g/m^3$).

Our group is exploring the question: How do fluctuations in particulate matter PM10 and PM2.5 predict respiratory-related Emergency Department visits across Virginia counties, and how do seasonal trends and socioeconomic factors moderate this relationship? This topic is worth exploring because respiratory illness poses many challenges to people's lives, and the quality of the air we breathe can change how often people end up in the Emergency Room. After the COVID-19 pandemic and with the recent flu season being one of the worst in recent history, Virginians are concerned about their risk of respiratory disease and are interested in preventative measures according to news reports from WWBT and WAVY. Looking at the amount of particulate matter in certain areas during certain weeks can help us predict the number of respiratory cases in the ER, help Virginians understand their health risks more clearly, and help hospitals prepare for busy weeks before they happen.

Specific stakeholders who would be interested in this include Virginia residents at risk of respiratory illness and healthcare providers who want to provide adequate care. Government representatives and the Virginia Department of Health could also use this to update safety protocols or make decisions about air pollution, while Emergency Departments can use it to be prepared for changes in the number of patients they need to treat.

Bibliography Annotation

We reviewed articles by Scott A. Weichenthal et al. and Jennifer L. Peel et al. regarding air pollution and ER visits. Both articles concluded that short-term fluctuations in air pollution consistently lead to more Emergency Room visits for respiratory issues, and both emphasized that children are especially vulnerable to these effects. Weichenthal found that PM2.5 showed the strongest associations when measured as a three-day mean concentration, rather than just a single day of exposure. Peel's study showed that the timing of these visits depends on the illness, with pneumonia and COPD cases appearing quickly (0–2 days), while asthma cases often peaked about a week after the pollution spike (5–8 days).

Several questions remain unanswered in the literature. We are still unsure about the most appropriate structure to use when linking pollution exposure to ER visits, and it is unclear how strongly PM2.5 and PM10 differ in their specific impact on respiratory outcomes. Additionally, the role of confounding seasonal effects, like winter weather and flu season intensity, needs further clarification. Other factors like geographic and demographic differences also need to be examined to identify exactly how vulnerable a specific population is, as well as whether health risks are driven by consistent air quality levels or short-term fluctuations.

Data

The data for this project comes from three primary government sources to ensure the information is legitimate and accurate. Respiratory health data was pulled from the Virginia Department of Health (VDH) via the Detailed Respiratory Emergency Department Visits Public Use Dataset. This includes weekly counts for COVID-19, Influenza, and RSV across different health districts from March 2020 to the present. Air quality data was sourced from the U.S. Environmental Protection Agency (EPA), specifically the 2024 Daily Data for PM10 and PM2.5 pollutants, which are recorded by monitors throughout Virginia. Finally, we pulled socioeconomic data, including Median Household Income and county rankings, from HDPulse (National Cancer Institute) and employment and wage data from the U.S. Bureau of Labor Statistics (BLS). All of these datasets are free, publicly available, and based on real-world observations rather than simulated data.

We originally wanted to use a lagged data structure for the particulate matter because research by Peel et al. (2005) suggests that pollution has a delayed effect on health, often hitting 2 to 5 days later. However, we could not perform a traditional lagged analysis because the VDH emergency room data is only available on a weekly basis, while the EPA data is daily. This mismatch in timing meant we had to stick to a contemporaneous weekly analysis instead of looking at day-by-day delays.

Cleaning the data required several steps to get these different sources to work together. First, we aggregated the daily EPA pollutant data into weekly averages to match the timeframe of the ER visit dataset. We also had to perform a lot of mapping to align the different geographic scales. We mapped the latitude and longitude of the air quality monitors and the specific cities and counties from the socioeconomic datasets onto the VDH's specific Health Districts. For the employment and wage data, we mapped the values specifically for January of each year in our dataset. The Median Household Income was already aggregated for 2019-2023 and was mapped to each row based on the specific city or county to ensure the socioeconomic context was accurate for each health district.

Visualizations

- Present the data visualizations that your group created in answering your questions of interest.

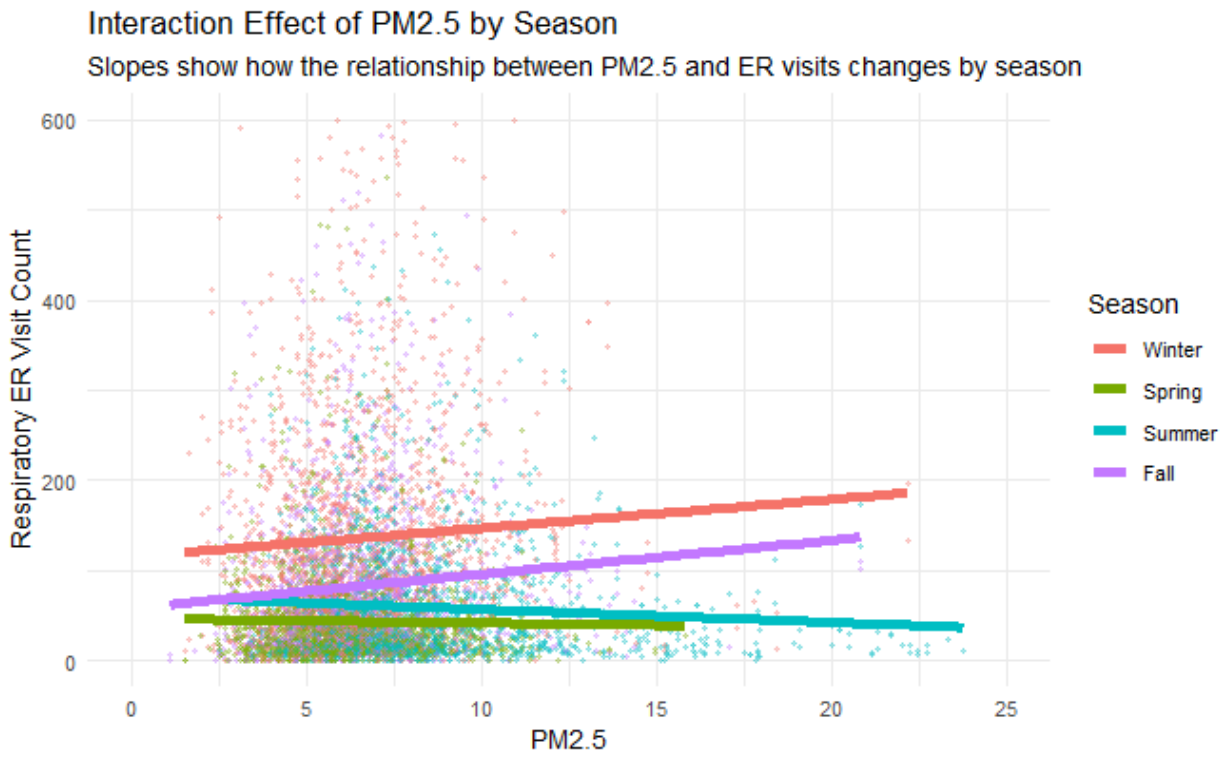


Figure 1: Interaction Effect of PM2.5 on Respiratory ER Visits by Season

Figure 1 highlights how the relationship between PM2.5 concentration and respiratory ER visits varies across seasons. While winter shows the highest baseline level of visits, indicated by the elevated intercept of the winter line, the slope for fall is noticeably steeper, meaning increases in PM2.5 lead to a larger rise in ER visits during this period. In contrast, summer shows a flat or slightly negative slope, suggesting that pollution has little impact when overall respiratory illness is low. This pattern supports the idea of a seasonal interaction, where the effect of air pollution is not constant but depends on underlying conditions like viral activity. These findings align with existing literature that emphasizes short-term pollution effects on respiratory health, while also extending it by showing that this impact is strongest during transitional periods like early fall. The implication is that public health responses should not use uniform pollution thresholds year-round, but instead adjust for seasonal sensitivity, particularly during fall when populations appear more vulnerable to pollution-related health effects.

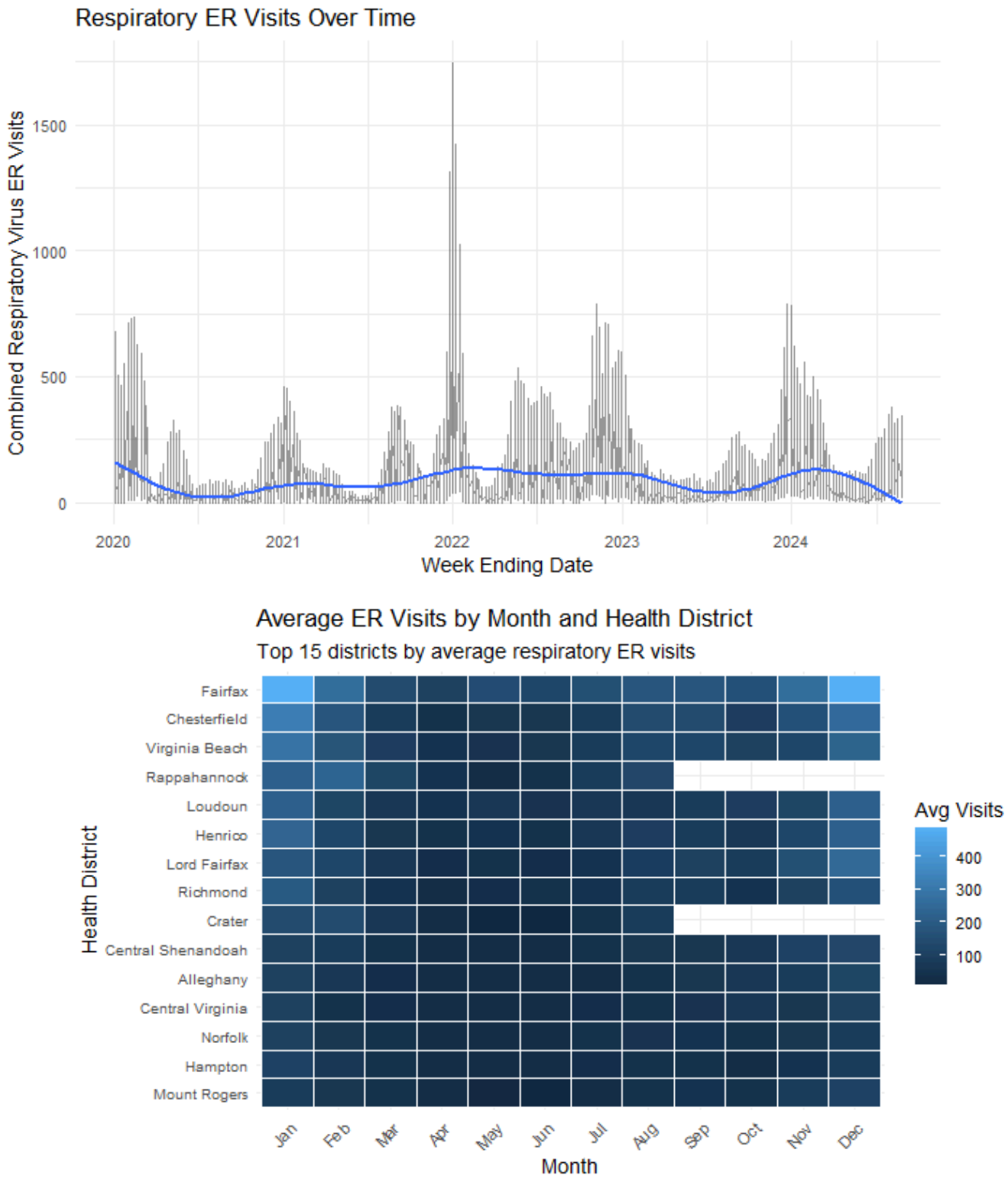


Figure 2 and 3: Respiratory ER Visits

These visualizations show that respiratory ER visits follow a strong seasonal pattern over time, with clear peaks in the winter months and lower levels during the summer. The heatmap further

reveals that this seasonal trend is consistent across the highest-volume health districts, with most districts experiencing darker (higher) values in winter and lighter (lower) values in spring and early summer. While there is some variation in overall levels between districts, the timing of increases and decreases is largely synchronized, suggesting that seasonal factors affect regions similarly. This supports the findings in Figure 1, where the impact of PM2.5 is not constant but depends on the time of year, particularly during the transition into the fall and winter respiratory season. Overall, these results reinforce existing literature that respiratory illness is driven by a combination of environmental and seasonal factors, and imply that public health planning should account for both geographic differences and predictable seasonal surges.

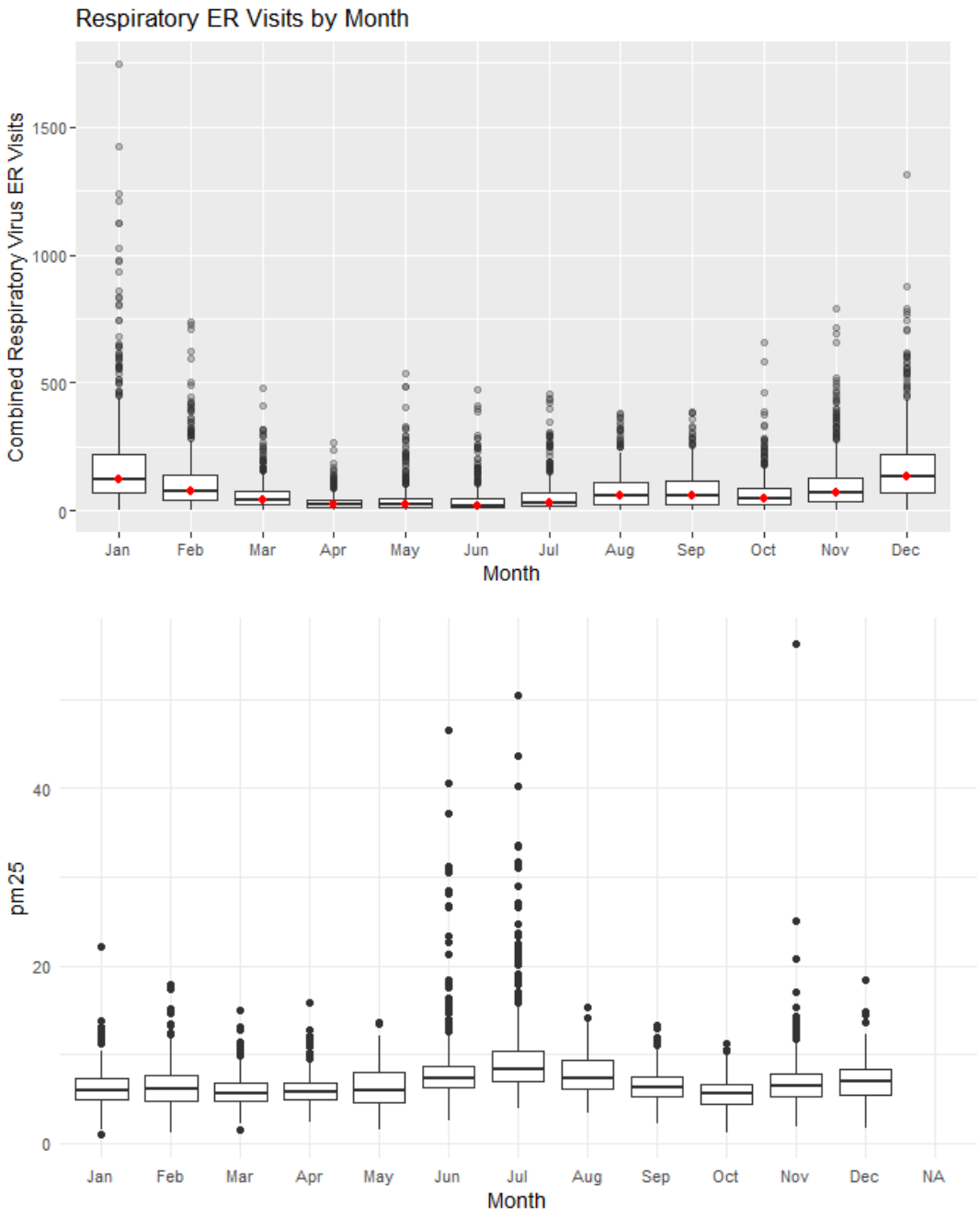


Figure 4 and 5:

Figures 4 and 5 together show that respiratory ER visits and PM2.5 follow different but related seasonal patterns. The top plot shows that ER visits are highest in winter (December and January) and lowest in spring and early summer, with a clear rise again in the fall. In contrast, the bottom plot shows that PM2.5 levels are more stable across the year, with only slight increases in the summer months and occasional spikes. This suggests that the strong seasonal pattern in ER visits is not driven solely by pollution levels, but also by underlying factors like viral activity and weather. However, because ER visits begin increasing in the fall while PM2.5 remains moderate, this supports the idea that populations may be more sensitive to pollution during certain seasons,

particularly in the fall. Overall, these visualizations reinforce that while PM2.5 contributes to respiratory outcomes, its impact depends heavily on seasonal conditions rather than just its absolute level.

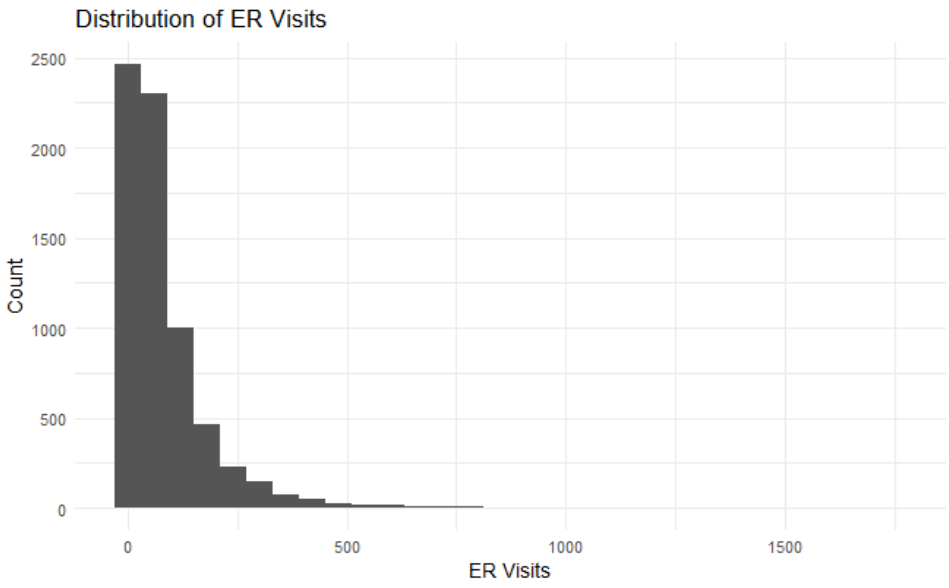


Figure 6:
This distribution is heavily right-skewed, with most ER visit counts clustered at lower values and a long tail of very high values. This means extreme spikes happen but are rare. The shape shows that the data are not normally distributed and have high variability, which explains why a Poisson model did not fit well and why a Negative Binomial model is more appropriate.

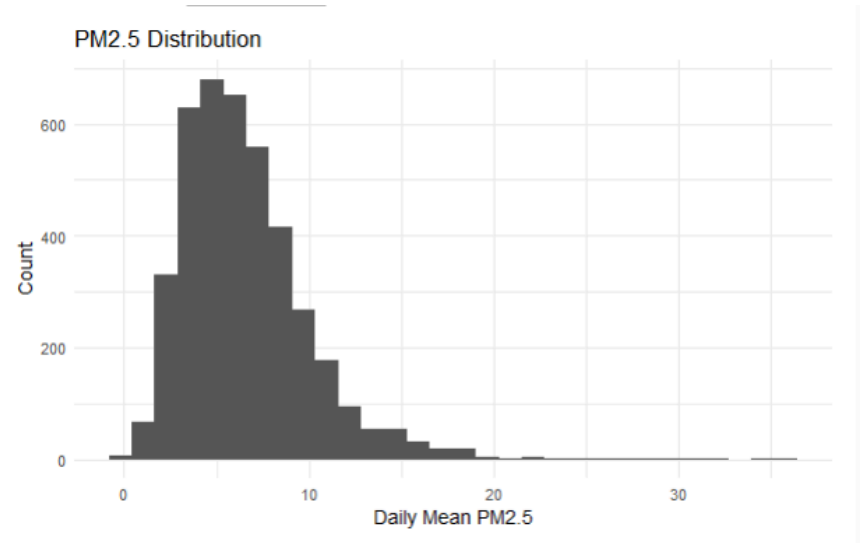


Figure 7:

This histogram shows that PM2.5 levels are right-skewed, meaning most observations are concentrated at lower values with a long tail of higher pollution levels. The majority of daily PM2.5 readings fall roughly between 2 and 10, indicating that typical air quality is relatively moderate. However, there are occasional higher values extending beyond 20 and even 30, which represent rare but significant pollution spikes. These extreme values are important because they can disproportionately impact respiratory health outcomes. Overall, the distribution suggests that while high pollution events are uncommon, they may play a key role in driving increases in ER visits.

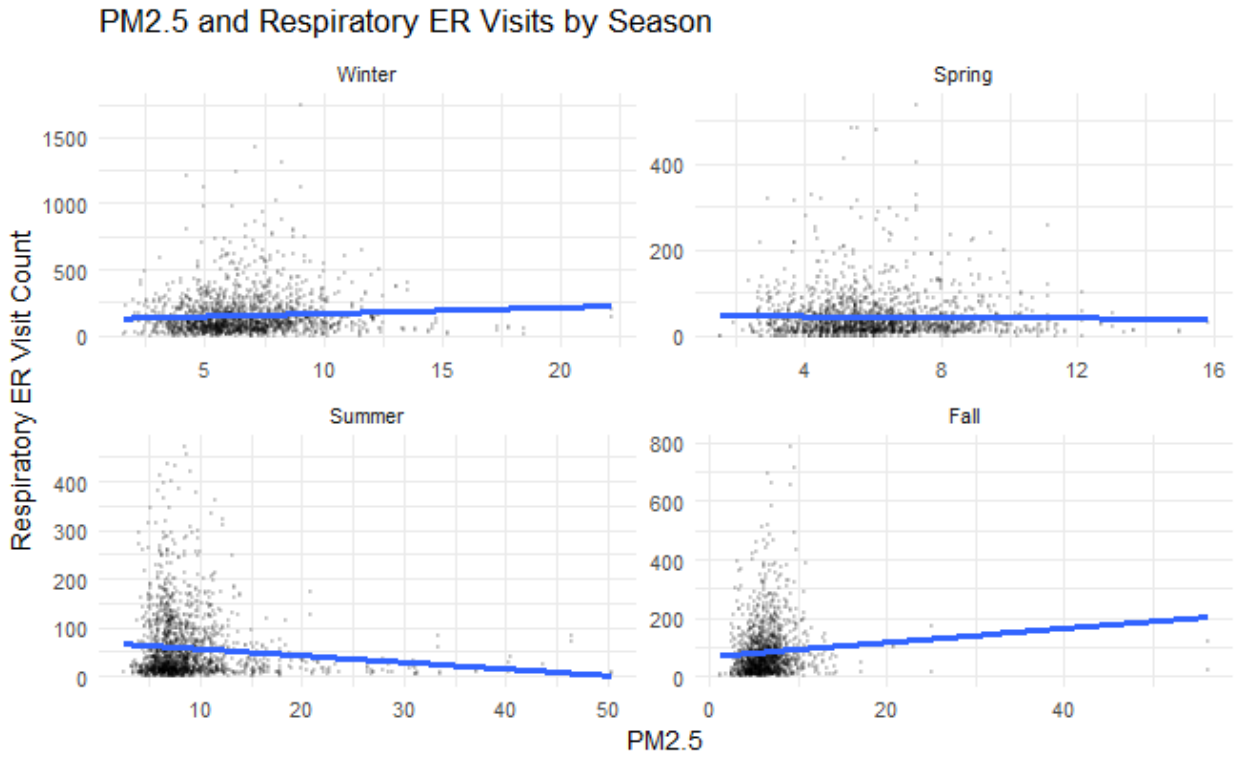


Figure 8:

Figure 8 shows that the relationship between PM2.5 and respiratory ER visits varies by season. Winter and fall display positive slopes, with fall showing the strongest increase, while summer shows a flat or slightly negative relationship. This indicates that PM2.5 has a greater impact on ER visits during fall and winter, supporting the idea of seasonal sensitivity to pollution.

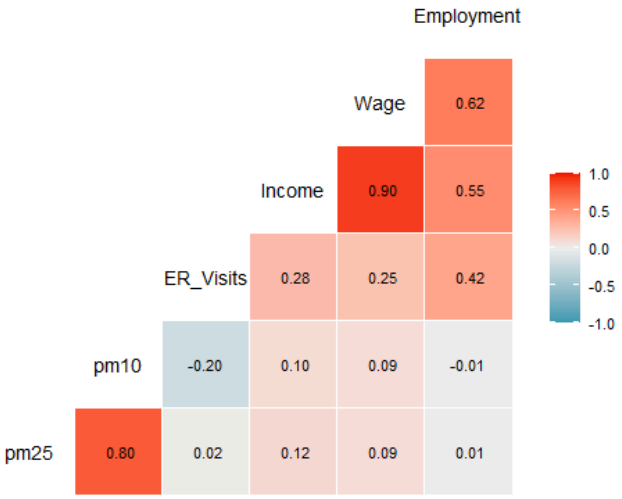


Figure 9:

This correlation matrix shows that PM2.5 and PM10 have very weak relationships with ER visits, with correlations close to zero, suggesting pollution alone does not strongly explain variation in respiratory outcomes. In contrast, ER visits are more moderately correlated with socioeconomic variables, especially employment (0.42), indicating these factors play a larger role in predicting healthcare demand. The strongest relationships in the matrix are between income, wage, and employment, confirming that these variables are closely related and likely capture similar underlying conditions like population density or access to care. Additionally, PM2.5 and PM10 are not strongly correlated with ER visits or with each other in a meaningful way for prediction. Overall, this supports your conclusion that respiratory ER visits are driven by a combination of seasonal and socioeconomic factors rather than pollution alone.

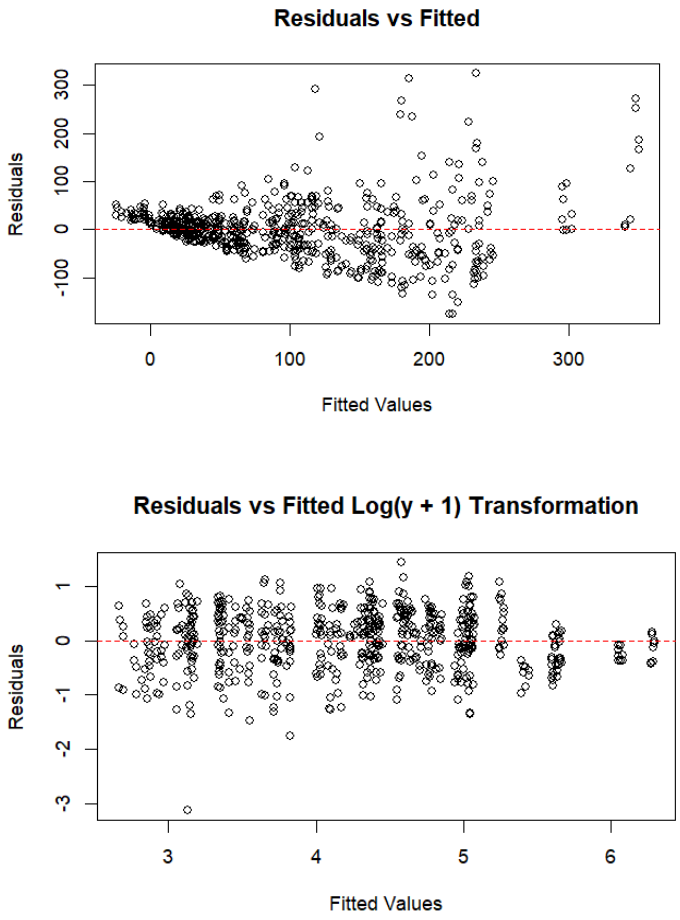


Figure 10 and 11:
Figure 10 shows that the residuals from the original linear model are not evenly spread, with variance increasing as fitted values increase. This funnel shape indicates heteroskedasticity, meaning the model violates the constant variance assumption and is not appropriate for this type of count data. In contrast, figure 11 plots using the log(y + 1) transformation shows a much more even spread of residuals around zero, suggesting improved variance stability. However, there are still some patterns and clustering, meaning the transformation helps but does not fully fix the issue. Overall, these plots justify moving away from a standard linear model and support the use of a Negative Binomial model, which better handles overdispersion in count data.

Methods

The analysis began with a multiple linear regression to model ER visit counts as a function of air quality and socioeconomic factors. While the initial model yielded an adjusted R^2 of 0.6245, it suffered from significant non-constant variance as shown by the heteroskedastic residuals fanning out, suggesting the linear model was inappropriate for count

data. We attempted a log(y + 1) transformation to address the skewness, which improved the adjusted R^2 to 0.6976, but the air quality variables remained statistically insignificant. Furthermore, a critical data limitation was discovered: PM10 was missing for 100% of observations in the final four months of the year. By switching to PM2.5, we secured near-complete temporal coverage.

To address the fundamental nature of the response variable as count data, we transitioned to a Poisson regression, specifically incorporating an interaction term between month and PM2.5 to allow the relationship to vary seasonally. This interaction improved the adjusted R^2 to 0.7282. However, the Poisson model was found to violate the assumption that the mean equals the variance, showing massive overdispersion. Consequently, we moved to a Negative Binomial regression.

A comparison of the two models using the Akaike Information Criterion (AIC) showed a staggering drop from 372,927.14 in the Poisson model to 61,538.84 in the Negative Binomial model, confirming the latter as the superior fit. We validated this choice using 5-fold cross-validation, which showed that while the Poisson model had a slightly lower average MSE, the Negative Binomial was significantly more stable, with a Standard Deviation of the MSE of 890.84 compared to the Poisson's 923.35.

Results

The final analysis reveals that respiratory ER visits are driven by a complex interplay of seasonality, socioeconomics, and air quality. The baseline relationship between PM2.5 and virus counts is positive in January (estimate of 2.072e-02), though the Negative Binomial model demonstrates that this relationship is highly dependent on the time of year. Specifically, we found significant positive interaction terms in the autumn, with September (6.823e-02) and October (7.225e-02) showing a heightened susceptibility to PM2.5 compared to the winter baseline. Conversely, summer months like July showed a strong negative interaction (-6.989e-02), suggesting the impact of pollution on ER visits is much less pronounced during the warmer months.

Socioeconomic factors played a dominant role in the model's predictive power, with Median Household Income showing a positive coefficient of 6.605e-06 and January Employment showing a massive z-value of 18.895, suggesting these are critical proxies for population density and healthcare access. These findings align with existing literature that suggests respiratory health outcomes are not just a result of pollution levels, but are heavily moderated by seasonal viral cycles. The successful shift from a Poisson to a Negative Binomial framework was essential for these conclusions; without accounting for the variance in ER counts, the model would have over-reported significance. This research suggests that public health interventions regarding air quality should be seasonally targeted, with a particular focus on the high-interaction months of September and October.

Interpretation

The final model indicates that the relationship between air pollution and health is not uniform throughout the year. While PM2.5 generally correlates with increased ER visits, this effect is most intense during September and October. This spike likely occurs because air quality fluctuations are intersecting with the start of the academic year and the early stages of the viral season. Surprisingly, the impact was significantly lower during the summer months, which contradicts the idea that pollution has a constant year-round effect. Furthermore, the high significance of Median Household Income and employment data suggests that socioeconomic status is a primary driver of ER visits. These variables likely act as proxies for healthcare access and population density, meaning that a county's economic profile is just as important as air quality when predicting hospital surges.

Capstone Experience

This project is distinctive because it successfully navigates several significant data challenges. We had to develop a specific method to align daily EPA pollutant measurements with weekly VDH hospital records, a hurdle that often prevents these two datasets from being used together. Additionally, when we discovered that PM10 data was missing for the final quarter of the year, we pivoted to PM2.5 to ensure the analysis remained temporally complete. Methodologically, we went beyond the standard linear approach. Using our advisor Dr. Jeffrey Woo's guidance, we recognized that linear models are not suited for count data containing zeros. By progressing through Poisson and ultimately settling on a Negative Binomial regression, we accounted for the overdispersion in the data to ensure our results were statistically valid. Managing these technical pivots and reconciling messy, real-world government datasets defined our capstone experience.

Appendix A (Data dictionary)

Variable	**Units/Classes (Quantitative/Categorical)**	**Description**
Week.Ending.Date	Quantitative, YYYY-MM-DD	Week-ending date for the week during which information was collected
Health.Region	Categorical	Health Region defined by the Virginia Department of Health
Health_District	Categorical	Health District defined by the Virginia Department of Health
Facility.Type	Categorical	Type of facility where visit data was collected
Year	Quantitative, year	Calendar year corresponding to the date of data collection
Week	Quantitative, week number	Week number corresponding to the date of data collection
Month	Quantitative, month number	Calendar month corresponding to the date of data collection
Percent.of.ED.Visits.Covid	Quantitative	The percent of total ER visits relating to COVID.
Percent.of.ED.Visits.Influenza	Quantitative	The percent of total ER visits relating to Influenza.
Percent.of.ED.Visits.RSV	Quantitative	The percent of total ER visits relating to RSV.
Percent.of.ED.Visits.Combined	Quantitative	The percent of total ER visits relating to COVID, Influenza, RSV, or all three (Combined). Separate columns for each disease.

Diagnosed.Covid.Count	Quantitative	The number of ER visits resulting in diagnoses given for COVID.
Diagnosed.Influenza.Count	Quantitative	The number of ER visits resulting in diagnoses given for Influenza.
Diagnosed.RSV.Count	Quantitative	The number of ER visits resulting in diagnoses given for RSV.
Combined.Respiratory.Viruses.Count	Quantitative	The number of ER visits resulting in diagnoses given for COVID, Influenza, RSV, or all three (Combined).
pm25	Quantitative	Concentration of PM2.5 detected by sensors in a particular location on a particular day
pm10	Quantitative	Concentration of PM10 ($\mu g/m^3$) detected by sensors in a particular location on a particular day
Median.Household.Income	Quantitative	Median household income for the health district.
Average.County.MHI.National.Ranking	Quantitative	National ranking of the district's median household income relative to all U.S. counties
Average.Weekly.Wage	Quantitative	Average weekly wage for the population of the Health District
January.Employment	Quantitative	Employment rate of the Health District for January of the given year

Appendix B (Sources)

Peel, J. L., Tolbert, P. E., Klein, M., Metzger, K. B., Flanders, W. D., Todd, K., Mulholland, J. A., Ryan, P. B., & Frumkin, H. (2005). Ambient air pollution and respiratory emergency department visits. *Epidemiology (Cambridge, Mass.)*, *16*(2), 164–174. https://doi.org/10.1097/01.ede.0000152905.42113.db

Weichenthal, S. A., Lavigne, E., Evans, G. J., Godri Pollitt, K. J., & Burnett, R. T. (2016). Fine Particulate Matter and Emergency Room Visits for Respiratory Illness. Effect Modification by Oxidative Potential. *American journal of respiratory and critical care medicine*, *194*(5), 577–586. https://doi.org/10.1164/rccm.201512-2434OC